

HARIRAMAKRRISHNAN RAMACHANDRAN

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PROFESSIONAL EXPERIENCE

SOFTWARE ENGINEER – HCL Technologies | Chennai, India

09/2021 – 01/2025 (3.5 Years)

- Designed and deployed **predictive analytics pipelines** using **Python (pandas, scikit-learn, NumPy)** and **SQL** to process enterprise transaction data across **PostgreSQL, MySQL, and Oracle** databases, enabling real-time operational insights for stakeholders and reducing data-processing latency by **40%**.
- Built interactive **Power BI** and **Tableau** dashboards with **DAX** measures and **Power Query** ETL logic to visualize SLA metrics, incident trends, and production-support KPIs across enterprise Java/.NET applications monitored via **Splunk** and **Dynatrace**, driving **25% improvement** in incident-response time.
- Developed **statistical analysis** and **regression models** in **R** and **Python** to forecast payment-transaction volumes and detect anomalies in **SWIFT messaging (MT/MX formats)** and **ISO 20022** message flows, supporting regulatory compliance and **99.8% transaction accuracy** in banking-operations environments.
- Engineered automated **data warehousing** solutions using **AWS (EC2, S3, Lambda)** and **Python** scripting to consolidate cross-system data from ERP platforms and operational business systems, reducing manual reporting overhead by **60%** and enabling enterprise-wide data governance.
- Implemented **machine learning classification models** using **scikit-learn** and **hypothesis testing** methodologies to segment banking clients and predict payment-reconciliation failures, achieving **88% model accuracy** and reducing post-implementation triage time by **35%**.
- Conducted **exploratory data analysis** and **A/B testing** on production datasets to validate feature impact on system reliability and business outcomes; communicated findings through **data visualization** dashboards and documented insights for cross-functional BA and operations teams in SEPA, cross-border payments, and PSD2-compliant environments.

SKILLS

Data Science & Analytics – Python, R, scikit-learn, pandas, NumPy, statistical analysis, regression, hypothesis testing, exploratory data analysis, A/B testing, machine learning, supervised learning, unsupervised learning, model evaluation metrics, time-series analysis.

Business Intelligence & Visualization – Power BI, Tableau, DAX, Power Query, data modelling, interactive dashboards, data warehousing, ETL pipeline design.

Databases & Data Engineering – SQL, PostgreSQL, MySQL, Oracle, schema design, ad-hoc query optimization, data validation, ETL automation, data-pipeline development.

Cloud & Infrastructure – AWS (EC2, S3, Lambda, CloudWatch), Docker, Linux, shell scripting, cron jobs, CI/CD (Jenkins, Git).

Domain Expertise – Banking & financial services, payments processing, SWIFT messaging (MT/MX), ISO 20022 message flows, SEPA, cross-border payments, PSD2 compliance, Open Banking, GDPR, data protection, ERP integration, change-control management, production support, incident management, SLA management.

Programming Languages & Web Frameworks – Python, Java, Spring Boot, REST APIs, Flask, FastAPI, React, Angular.

Tools & Platforms – Jira, Confluence, Splunk, Dynatrace, Git, Jenkins, Agile/Scrum.

EDUCATION AND TRAINING

PROJECTS

Real-Time Payments Analytics & Fraud Detection Platform (HCL Technologies)

- Architected end-to-end **data science pipeline** ingesting **ISO 20022 message flows** and **SWIFT transaction data** via **AWS Lambda** into **S3 data lake**, then processing via **Python (pandas, scikit-learn)** to detect anomalies in cross-border and SEPA payment patterns, reducing fraud-detection latency from **6 hours to 12 minutes**.
- Built **predictive machine-learning models** using **supervised classification** and **unsupervised clustering** to segment high-risk payment corridors and flag reconciliation failures before settlement; achieved **91% precision** and **87% recall** in classifying fraudulent transaction sequences.
- Developed interactive **Power BI dashboards** with **DAX** measures and real-time **Power Query** refresh logic to surface transaction-flow visibility, payment-status metrics, and regulatory-compliance KPIs for banking operations and compliance teams; reduced **UAT support queries by 55%**.
- Engineered **data warehousing** layer on **PostgreSQL** and **Oracle** to consolidate transaction master data across ERP systems, SWIFT messaging platforms, and tokenization services; implemented **automated ETL jobs** via **Python scripting** and **cron** to refresh data daily, supporting **99.9% audit-trail accuracy** for GDPR compliance.
- Conducted **statistical analysis** and **A/B testing** on payment-routing rules and reconciliation workflows, using **R** for hypothesis validation and **regression modeling** to forecast transaction volumes; recommendations reduced settlement exceptions by **32%** across **45+ payment corridors**.

Enterprise Business System Data Integration & BI Consolidation (HCL Technologies)

- Designed **cross-system data integration** connecting ERP platforms, banking-operations dashboards, and business-system databases via **REST APIs** and **SQL** data flows; **Python** automation scripts managed **change-control** and **UAT coordination** across **8+ interconnected systems**, reducing post-go-live triage incidents by **48%**.
- Created unified **Tableau dashboards** and **Power BI reports** consolidating KPIs from ERP modules, production-support metrics (**Splunk, Dynatrace**), and operational systems; implemented **data modelling** layer in **PostgreSQL** to support self-service analytics for business-system users, enabling **20% faster decision cycles**.
- Automated **data-quality validation** and **ETL reconciliation workflows** using **Python** and **SQL** to ensure consistency across ERP, banking, and tokenization platforms; maintained **99.95% data accuracy** during system-configuration updates and change-management cycles.

ACHIEVEMENTS & CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California, Davis** (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)